Two Phase Bipolar Stepper Motor Controller

PCB Design Project | Tool: KiCad | 2-Layer PCB

Design Tool	KiCad
Board Type	2-Layer PCB
Power Supply	VSS up to 46V, Logic 5V

1. Project Overview

2-phase bipolar stepper motor driver PCB using L297+L298N ICs. Supports STEP/DIR input, full-bridge drive up to 2A/channel, with flyback diode protection. Designed in KiCad with Gerber output.

2. Key Components

- L297 Stepper Controller IC
- L298N Dual Full-Bridge Driver
- Schottky Diodes (8x) – Flyback protection
- Screw Terminal Blocks – Motor/Power connectors
- Capacitors – Decoupling/Filtering
- Current Sense Resistors

3. Design Features

- Full-bridge bipolar drive for 2-phase stepper motor
- L297 generates STEP/DIR phase sequences automatically
- L298N handles up to 2A per channel (4A peak)
- Flyback diodes protect against inductive voltage spikes
- Screw terminals for easy motor and power wiring
- Gerber files generated for PCB fabrication

4. Signal / Functional Flow

[1] MCU / Signal Source ↓
[2] STEP / DIR Inputs ↓
[3] L297 – Phase Sequencer ↓
[4] L298N – H-Bridge Driver ↓
[5] Stepper Motor Coils ↓
[6] Flyback Diodes ← Inductive Kickback Protection

5. Resume / Portfolio Description

- Designed 2-phase bipolar stepper motor controller PCB in KiCad using L297 stepper IC + L298N H-bridge driver with flyback diode protection.
- Generated complete Gerber files; board supports STEP/DIR interface, 2A/ch drive current, and screw terminal I/O connections.

6. Short Description (Portfolio / GitHub – ≤350 chars)

2-phase bipolar stepper motor driver PCB using L297+L298N ICs. Supports STEP/DIR input, full-bridge drive up to 2A/channel, with flyback diode protection. Designed in KiCad with Gerber output.

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